VMware ESXi
HYPERVISOR

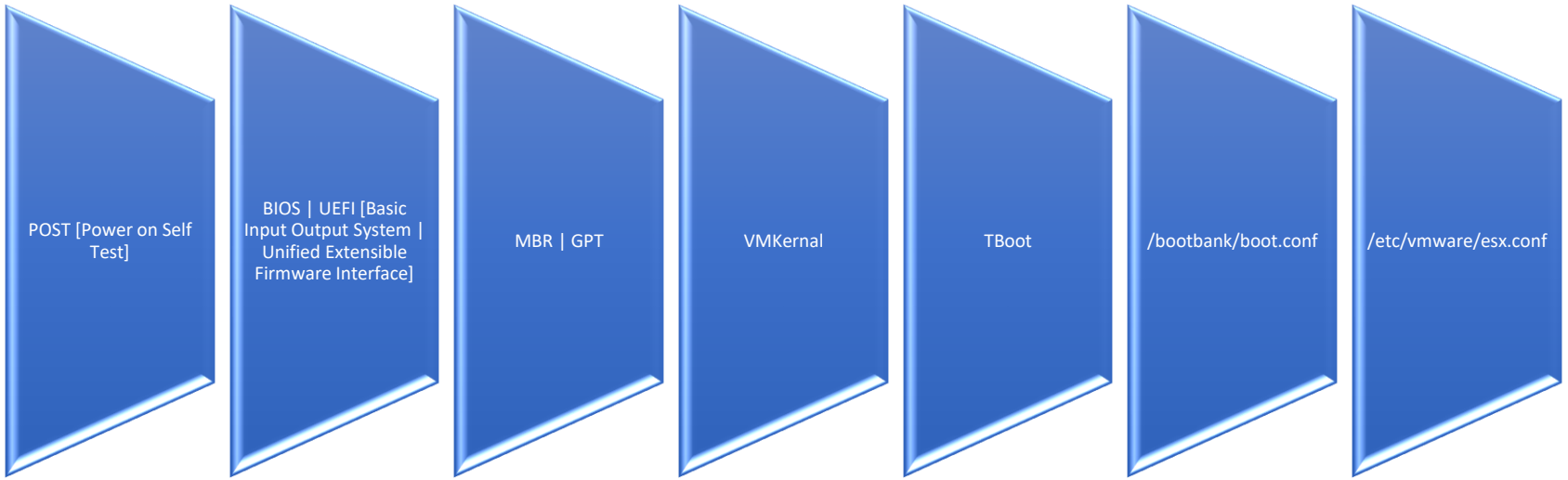
BOOTING PROCESS OF VMware ESXi 8.0.U3



VMware ESXi HYPERVISOR

BOOTING PROCESS OF VMware ESXi 8.0





VMkernel is a POSIX (Portable Operating System Interface)-like operating system (OS) developed by VMware.

To provide display in VMware ESXi.

To load all device & driver's.

To load DCUI in VMware ESXi Hypervisor

VMware ESXi
HYPERVISOR

BOOTING PROCESS OF VMware ESXi 8.0



Master Boot Record (MBR)

Master Boot Record (MBR) disks use the standard BIOS partition table. GUID partition table (GPT) disks use the Unified Extensible Firmware Interface (UEFI). One advantage of GPT disks is that you can have more than four partitions on each disk. GPT is also required for disks larger than 2 terabytes (TB).

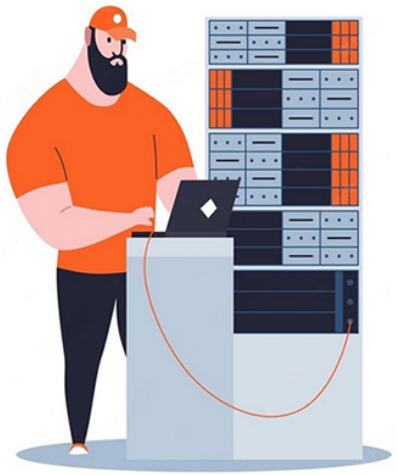
MBR is the older, more compatible partitioning scheme, suitable for smaller disks and legacy systems, but it has limitations in terms of partition count, disk size, and reliability.

GUID Partition Table (GPT)

The GUID Partition Table (GPT) was introduced as part of the Unified Extensible Firmware Interface (UEFI) initiative. GPT provides a more flexible mechanism for partitioning disks than the older Master Boot Record (MBR) partitioning scheme that was common to PCs.

GPT is the modern, more robust partitioning scheme that offers support for larger disks, more partitions, and improved data integrity. It is required for booting with UEFI firmware and is the recommended choice for new systems, especially those with large drives or modern operating systems.

Feature	MBR	GPT
Partition Limit	4 primary (or 3 primary + 1 extended)	Up to 128 partitions
Disk Size Limit	2 TB	9.4 zettabytes (almost unlimited)
Boot Process	BIOS-based	UEFI-based
Data Integrity	No redundancy or checksums	CRC32 checksums and redundancy
OS Compatibility	Compatible with most OS, including older	Requires UEFI and modern OS (Windows 7+, Linux)
Redundancy	No redundancy	Backup copy of partition table at disk's end
Boot Mode	BIOS only	UEFI (no BIOS support)
Security Features	No secure boot	Secure boot supported by UEFI



VMware ESXi
HYPERVISOR

SERVICES OF VMware ESXi 8.0

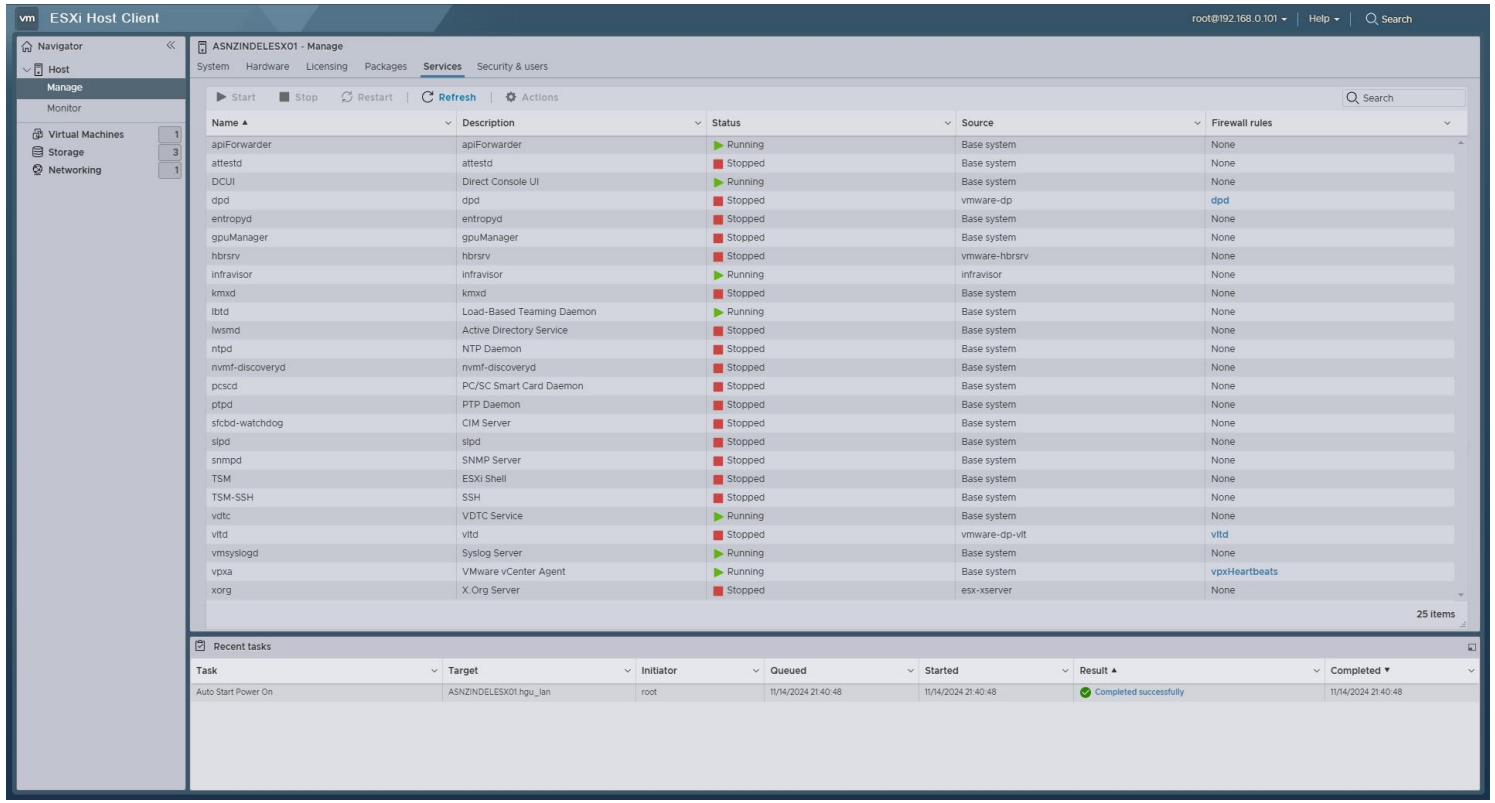


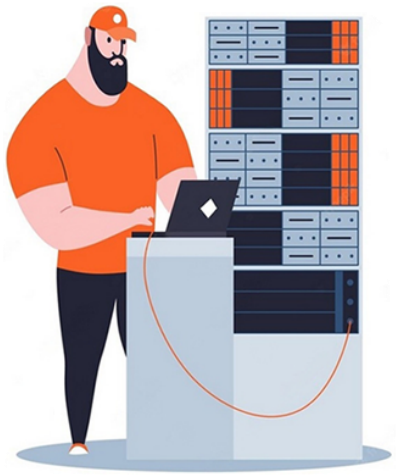
VMware ESXi
HYPERVISOR

SERVICES OF VMware ESXi 8.0

There are 25 Hypervisor Services in VMware ESXi 8.0.U3

- SNMP Server
- PC/SC Smart Card Daemon
- Load-Based Teaming Daemon
- ESXi Shell
- X.Org Server
- VMware vCenter Agent
- NTP Daemon
- Active Directory Service
- VDTC Service
- SSH
- Syslog Server
- Direct Console UI
- CIM Server
- SLPD
- PTP Daemon
- ENTROPYD
- ATTESTD
- **KMXD**
- **VLTD**
- **API Forwarder**
- **DPD**
- **GPU Manager**
- **INFRAVISOR**
- **HBRSRV**
- **NVMF-DISCOVERYD**

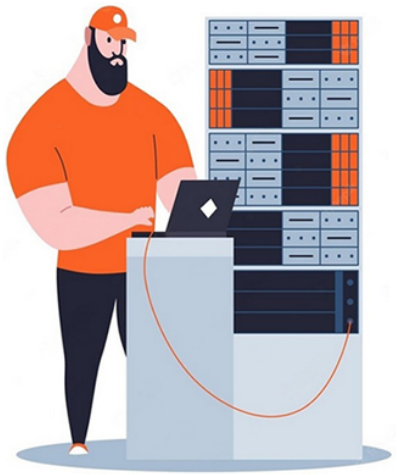




SERVICES OF VMware ESXi 8.0



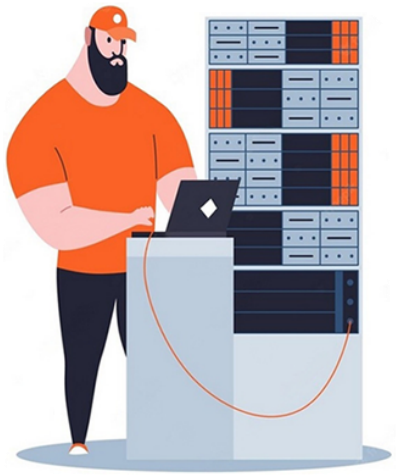
- **SNMP Server**
  - SNMP service provide logs to export in operating system such as Windows / Linux.
- **PC/SC Smart Card Daemon**
  - The PC/SC daemon is used to manage connections to PC and SC smart card readers. Although PC/SC is usually a low priority task, it can often use more CPU than any other daemon. This additional background noise can lead to higher pre-emption costs to real-time tasks and other undesirable impacts on determinism.
- **Load-Based Teaming Daemon**
  - LBTD service is responsible for NIC Teaming.
- **ESXi Shell**
  - This service provide you shell platform in ESXi.
- **X.Org Server**
  - XORG service is responsible for providing 3D graphics service for guest operating system.
- **VMware vCenter Agent**
  - This service is responsible to connect your ESXi with vCenter server.
- **NTP Daemon**
  - NTPD service is responsible for time. If you want to create time server so VMware is providing you this service. Activate this service on your host. This service work on port no 123 with UDP protocol.
- **Active Directory Service**
  - This service provide Active Directory authentication as well as domain user can login on ESXi host as per given permission.
- **VDTC Service**
  - VMware vSphere Distributed Tracing Collector.
- **SSH**
  - SSH service give you remote management of ESXi using putty or any other command line software. This service work on port no 22 with TCP.



SERVICES OF VMware ESXi 8.0



- **Syslog Server**
  - Syslog service is responsible to generate all type of ESXi logs.
- **Direct Console UI**
  - This service is responsible to load your DCUI in ESXi host.
- **CIM Server**
  - Use only for developers to customization of ESXi host.
- **SLPD**
  - SLPD service is used by the vSphere client to discover hardware inventory on your hosts.
- **PTP Daemon**
  - The Precision Time Protocol allows both software- and hardware-based timestamping on an ESXi hosts and provides a highly accurate time synchronization. Starting with vSphere 7.0 Update 3, PTP provisions hardware-based timestamping for the virtual machines and the hosts within a network.
  - The PTP and the NTP services cannot run simultaneously. Stop the NTP service and then enable the PTP service. Additionally, when you enable the PTP service, the manual time configuration becomes inactive.
- **ENTROPYD**
  - In computing, the term “entropy” refers to random characters and data that are collected for use in cryptography, such as generating encryption keys to secure data transmitted over a network. Entropy is required by security for generating keys and communicating securely over the network. Entropy is often collected from a variety of sources on a system.
- **ATTESTD**
  - Attestation Service the state of the ESXi hosts using a Trusted Platform Module (TPM) 2.0 chip for software measurement and reporting. The TPM measures the software stack and configuration data and reports it to the Attestation service.



SERVICES OF VMware ESXi 8.0



- **API Forwarder**
- **DPD**
- **GPUManager**
- **INFRAVISOR**
- **HBRSRV**
- **NVMF-DISCOVERYD**
- **VLTD**
- **KMXD**

